

University of Wisconsin–Madison

All field types are compatible with both SQLite (development) and MySQL (production) via Django .

Models Covered

- ☐ User — All roles: student, faculty, admin, staff, parent, prospective
- ☐ School / College — All 13 UW-Madison schools and colleges
- ☐ Department — 160+ academic departments
- ☐ Program — Undergraduate majors, graduate programs, certificates
- ☐ Course — Full 9,000+ course catalog
- ☐ CourseSection — Individual sections with instructor & scheduling
- ☐ Faculty — Faculty profiles linked to user accounts
- ☐ Enrollment — Student-section enrollment records
- ☐ Application — Admissions applications for all types
- ☐ StudyAbroadProgram — 60+ country programs
- ☐ ResearchOpportunity — UG and Graduate research positions
- ☐ Event — Academic, sports, cultural, career events
- ☐ Library — Campus library locations and resources
- ☐ SportsTeam — club sports teams
- ☐ Notification — User notification/alert system

1. User Model (loki team)

Central authentication and profile model for all platform users. Extend Django's AbstractUser or use a custom model. Covers students, faculty, admin, staff, parents, and prospective students.

Field Name	Django Field Type	DB Column / Notes	Description / Example
user_id	AutoField (PK)	<i>PRIMARY KEY</i>	Auto-incremented unique user identifier
username	CharField(150)	<i>UNIQUE, NOT NULL</i>	Unique login username
first_name	CharField(100)	<i>NOT NULL</i>	User's first name
last_name	CharField(100)	<i>NOT NULL</i>	User's last name
email	EmailField(254)	<i>UNIQUE, NOT NULL</i>	Primary contact email
password_hash	CharField(255)	<i>NOT NULL</i>	Hashed password (Django default)

role	CharField(30)	<i>choices, NOT NULL</i>	student / faculty / admin / staff / parent / prospective
profile_photo_url	URLField(500)	<i>nullable</i>	Link to uploaded profile picture
phone_number	CharField(20)	<i>nullable</i>	Contact phone number
date_of_birth	DateField()	<i>nullable</i>	User's date of birth
gender	CharField(20)	<i>nullable, choices</i>	Male, Female, Trans, LGBTQ+ prefer not to say, Other .
is_active	BooleanField()	<i>default=True</i>	Account active status
is_verified	BooleanField()	<i>default=False</i>	Email verification status
created_at	DateTimeField()	<i>auto_now_add=True</i>	Account creation timestamp
last_login	DateTimeField()	<i>nullable</i>	Last successful login timestamp
address_line1	CharField(255)	<i>nullable</i>	Street address line 1
city	CharField(100)	<i>nullable</i>	City of residence
state	CharField(100)	<i>nullable</i>	State of residence
zip_code	CharField(10)	<i>nullable</i>	Postal ZIP code
country	CharField(100)	<i>default='USA'</i>	Country of residence
notification_pref	JSONField()	<i>default=dict</i>	Email/SMS/push notification preferences
ssn	OneToOneField → SSNRecord	Yes/Yes	FK to SSNRecord model
guardian1	ForeignKey → Guardian	Yes/Yes	Primary guardian
guardian2	ForeignKey → Guardian	Yes/Yes	Secondary guardian (supports same-sex parents)

Use `AUTH_USER_MODEL = 'myapp.User'` in `settings.py` and extend `AbstractBaseUser` or `AbstractUser` for full control.

2. School / College Model (Steve Team)

Represents each of the 13 schools and colleges under UW-Madison (e.g., College of Engineering, School of Business).

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>school_id</code>	AutoField (PK)	<i>PRIMARY KEY</i>	Auto-incremented school identifier
<code>name</code>	CharField(200)	<i>UNIQUE, NOT NULL</i>	e.g. School of Engineering
<code>abbreviation</code>	CharField(20)	<i>UNIQUE</i>	e.g. SOE, COL, SOM
<code>dean_name</code>	CharField(150)	<i>nullable</i>	Full name of the current dean
<code>dean_email</code>	EmailField(254)	<i>nullable</i>	Dean's official email address
<code>website_url</code>	URLField(500)	<i>nullable</i>	Official school/college website
<code>logo_url</code>	URLField(500)	<i>nullable</i>	URL to the school logo image
<code>description</code>	TextField()	<i>nullable</i>	Overview/mission of the school
<code>established_year</code>	IntegerField()	<i>nullable</i>	Year the school was established
<code>contact_phone</code>	CharField(20)	<i>nullable</i>	Main office phone number
<code>location_building</code>	CharField(200)	<i>nullable</i>	Main building name/address
<code>is_active</code>	BooleanField()	<i>default=True</i>	Whether school is currently active
<code>created_at</code>	DateTimeField()	<i>auto_now_add=True</i>	Record creation timestamp

3. Department Model (Bela Team)

Academic departments (160+) nested under schools. Each department hosts courses, programs, and faculty.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>dept_id</code>	AutoField (PK)	<i>PRIMARY KEY</i>	Unique department identifier
<code>name</code>	CharField(200)	<i>NOT NULL</i>	Full department name
<code>school</code>	ForeignKey(School)	<i>ON DELETE CASCADE</i>	Parent school/college FK
<code>head_name</code>	CharField(150)	<i>nullable</i>	Department head/chair name
<code>head_email</code>	EmailField(254)	<i>nullable</i>	Department head email
<code>website_url</code>	URLField(500)	<i>nullable</i>	Department website URL
<code>description</code>	TextField()	<i>nullable</i>	Department overview
<code>office_location</code>	CharField(200)	<i>nullable</i>	Building/room number
<code>contact_phone</code>	CharField(20)	<i>nullable</i>	Department phone number
<code>contact_email</code>	EmailField(254)	<i>nullable</i>	Department general email
<code>is_active</code>	BooleanField()	<i>default=True</i>	Active status
<code>created_at</code>	DateTimeField()	<i>auto_now_add=True</i>	Record creation timestamp

4. Program Model

Degree programs including undergraduate majors (136+), graduate programs (279+), and certificates. Linked to school and department .

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>program_id</code>	AutoField (PK)	<i>PRIMARY KEY</i>	Unique program identifier
<code>name</code>	CharField(200)	<i>NOT NULL</i>	Program name (e.g. Computer Science)
<code>code</code>	CharField(20)	<i>UNIQUE</i>	Short program code
<code>type</code>	CharField(30)	<i>choices</i>	Major / Minor / Certificate / Graduate
<code>degree_level</code>	CharField(20)	<i>choices</i>	UG / Masters / PhD / Professional
<code>school</code>	ForeignKey(School)	<i>ON DELETE CASCADE</i>	Associated school FK
<code>department</code>	ForeignKey(Department)	<i>nullable FK</i>	Primary department offering the program
<code>description</code>	TextField()	<i>nullable</i>	Full program description
<code>total_credits</code>	IntegerField()	<i>NOT NULL</i>	Total credit hours required
<code>duration_years</code>	FloatField()	<i>nullable</i>	Typical years to complete (e.g. 4.0)
<code>admission_requirements</code>	TextField()	<i>nullable</i>	Requirements for admission
<code>career_outcomes</code>	JSONField()	<i>default=list</i>	List of common career paths
<code>is_online</code>	BooleanField()	<i>default=False</i>	Available fully online
<code>application_deadline</code>	DateField()	<i>nullable</i>	Application deadline date
<code>tuition_per_credit</code>	DecimalField(8,2)	<i>nullable</i>	Cost per credit hour

<code>is_active</code>	<code>BooleanField()</code>	<code>default=True</code>	Program active status
<code>created_at</code>	<code>DateTimeField()</code>	<code>auto_now_add=True</code>	Record creation timestamp

5. Course Model

Catalog of all 9,000+ courses offered at UW-Madison. Each course belongs to a department and can have prerequisite courses.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>course_id</code>	<code>AutoField (PK)</code>	<i>PRIMARY KEY</i>	Unique course identifier
<code>code</code>	<code>CharField(20)</code>	<i>UNIQUE</i>	e.g. CS 540, MATH 221
<code>title</code>	<code>CharField(300)</code>	<i>NOT NULL</i>	Full course title
<code>description</code>	<code>TextField()</code>	<i>nullable</i>	Course description/overview
<code>credits</code>	<code>IntegerField()</code>	<i>NOT NULL</i>	Credit hours (1-6 typically)
<code>department</code>	<code>ForeignKey(Department)</code>	<i>ON DELETE SET NULL</i>	Offering department FK
<code>level</code>	<code>CharField(20)</code>	<i>choices</i>	UG / Grad / Both
<code>prerequisites</code>	<code>ManyToManyField(Course)</code>	<i>blank=True</i>	Self-referential prerequisite courses
<code>syllabus_url</code>	<code>URLField(500)</code>	<i>nullable</i>	Link to course syllabus PDF
<code>max_enrollment</code>	<code>IntegerField()</code>	<i>nullable</i>	Maximum seats per section
<code>is_active</code>	<code>BooleanField()</code>	<code>default=True</code>	Course offered this year
<code>created_at</code>	<code>DateTimeField()</code>	<code>auto_now_add=True</code>	Record creation timestamp

6. CourseSection Model

Individual sections of a course for a specific semester, with assigned faculty, room, schedule, and capacity tracking.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>section_id</code>	AutoField (PK)	<i>PRIMARY KEY</i>	Unique section identifier
<code>course</code>	ForeignKey(Course)	<i>ON DELETE CASCADE</i>	Parent course FK
<code>section_number</code>	CharField(10)	<i>NOT NULL</i>	e.g. 001, 002, LEC
<code>semester</code>	CharField(20)	<i>choices</i>	Summer / Winter
<code>year</code>	IntegerField()	<i>NOT NULL</i>	Academic year (e.g. 2025)
<code>faculty</code>	ForeignKey(Faculty)	<i>nullable FK</i>	Assigned instructor FK
<code>schedule_days</code>	CharField(50)	<i>nullable</i>	
<code>start_time</code>	TimeField()	<i>nullable</i>	Class start time
<code>end_time</code>	TimeField()	<i>nullable</i>	Class end time
<code>classroom</code>	CharField(100)	<i>nullable</i>	Building and room (e.g. CS 1240)
<code>delivery_mode</code>	CharField(30)	<i>choices</i>	In-person / Online / Hybrid
<code>capacity</code>	IntegerField()	<i>NOT NULL</i>	Seat capacity for this section
<code>enrolled_count</code>	IntegerField()	<i>default=0</i>	Current enrolled students count
<code>waitlist_count</code>	IntegerField()	<i>default=0</i>	Current waitlist count

<code>is_open</code>	<code>BooleanField()</code>	<i>default=True</i>	Section open for enrollment
<code>created_at</code>	<code>DateTimeField()</code>	<i>auto_now_add=True</i>	Record creation timestamp

7. Faculty Model

Faculty profiles linked one-to-one with the User model. Includes academic rank, department, office details, and research specialization.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>faculty_id</code>	<code>AutoField (PK)</code>	<i>PRIMARY KEY</i>	Unique faculty identifier
<code>user</code>	<code>OneToOneField(User)</code>	<i>ON DELETE CASCADE</i>	Linked User account FK
<code>employee_id</code>	<code>CharField(50)</code>	<i>UNIQUE</i>	University employee ID
<code>title</code>	<code>CharField(50)</code>	<i>choices</i>	Professor / Assoc. Prof / Asst. Prof / Lecturer
<code>department</code>	<code>ForeignKey(Department)</code>	<i>ON DELETE SET NULL</i>	Primary department FK
<code>specialization</code>	<code>CharField(300)</code>	<i>nullable</i>	Research/teaching specialization
<code>bio</code>	<code>TextField()</code>	<i>nullable</i>	Faculty biography
<code>office_location</code>	<code>CharField(200)</code>	<i>nullable</i>	Office building and room
<code>office_hours</code>	<code>CharField(300)</code>	<i>nullable</i>	e.g. Mon 2-4pm, Wed 10-12pm
<code>personal_website</code>	<code>URLField(500)</code>	<i>nullable</i>	Personal or lab website URL
<code>hire_date</code>	<code>DateField()</code>	<i>nullable</i>	Date hired by university
<code>is_active</code>	<code>BooleanField()</code>	<i>default=True</i>	Currently employed

<code>created_at</code>	<code>DateTimeField()</code>	<code>auto_now_add=True</code>	Record creation timestamp
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8. Enrollment Model

Records each student's enrollment in a CourseSection. Tracks grade, status, attendance Used by Student Dashboard.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>enrollment_id</code>	<code>AutoField (PK)</code>	<i>PRIMARY KEY</i>	Unique enrollment record ID
<code>student</code>	<code>ForeignKey(User)</code>	<i>ON DELETE CASCADE</i>	Enrolled student FK
<code>section</code>	<code>ForeignKey(CourseSection)</code>	<i>ON DELETE CASCADE</i>	Enrolled section FK
<code>status</code>	<code>CharField(30)</code>	<i>choices</i>	Enrolled / Waitlisted / Dropped / Completed
<code>grade</code>	<code>CharField(5)</code>	<i>nullable</i>	Final grade (A, AB, B, BC)
<code>grade_points</code>	<code>FloatField()</code>	<i>nullable</i>	Grade points for GPA calculation
<code>enrollment_date</code>	<code>DateTimeField()</code>	<code>auto_now_add=True</code>	Date/time of enrollment
<code>drop_date</code>	<code>DateTimeField()</code>	<i>nullable</i>	Date/time if student dropped
<code>attendance_pct</code>	<code>FloatField()</code>	<i>nullable</i>	Attendance percentage (0.0 - 100.0)
<code>updated_at</code>	<code>DateTimeField()</code>	<code>auto_now=True</code>	Last update timestamp

9. Application Model

Admissions applications from prospective students for undergraduate, transfer, graduate, and international programs.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>application_id</code>	AutoField (PK)	<i>PRIMARY KEY</i>	Unique application identifier
<code>applicant</code>	ForeignKey(User)	<i>ON DELETE CASCADE</i>	Applying user FK
<code>program</code>	ForeignKey(Program)	<i>ON DELETE CASCADE</i>	Applied program FK
<code>application_type</code>	CharField(30)	<i>choices</i>	First-year / Transfer / Graduate / International
<code>status</code>	CharField(30)	<i>choices</i>	Draft / Submitted / Under Review / Accepted / Rejected / Waitlisted
<code>semester_intended</code>	CharField(20)	<i>choices</i>	Fall / Spring / Summer
<code>year_intended</code>	IntegerField()	<i>NOT NULL</i>	Intended enrollment year
<code>gpa</code>	FloatField()	<i>nullable</i>	Reported GPA (0.0 - 4.0)
<code>test_score_type</code>	CharField(20)	<i>nullable</i>	SAT / ACT / GRE / GMAT / None

			SAT — <i>Scholastic Assessment Test</i> ACT — <i>American College Testing</i> GRE — <i>Graduate Record Examination</i> GMAT — <i>Graduate Management Admission Test</i>
<code>test_score</code>	<code>IntegerField()</code>	<i>nullable</i>	Standardized test score
<code>personal_statement</code>	<code>TextField()</code>	<i>nullable</i>	Applicant's personal statement
<code>submitted_at</code>	<code>DateTimeField()</code>	<i>nullable</i>	Submission timestamp
<code>decision_at</code>	<code>DateTimeField()</code>	<i>nullable</i>	Decision date/time
<code>financial_aid_requested</code>	<code>BooleanField()</code>	<i>default=False</i>	Aid requested flag
<code>created_at</code>	<code>DateTimeField()</code>	<i>auto_now_add=True</i>	Application record creation

10. Study Abroad Program Model

Represents over 60 countries' study abroad programs available to UW students, including partner institutions and cost estimates.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>program_id</code>	<code>AutoField (PK)</code>	<i>PRIMARY KEY</i>	Unique study abroad program ID
<code>name</code>	<code>CharField(300)</code>	<i>NOT NULL</i>	Program name
<code>country</code>	<code>CharField(100)</code>	<i>NOT NULL</i>	Host country
<code>city</code>	<code>CharField(100)</code>	<i>nullable</i>	Host city
<code>partner_institution</code>	<code>CharField(200)</code>	<i>nullable</i>	Partner university name
<code>description</code>	<code>TextField()</code>	<i>nullable</i>	Program description
<code>duration_weeks</code>	<code>IntegerField()</code>	<i>nullable</i>	Length in weeks

<code>semester</code>	CharField(20)	<i>choices</i>	Fall / Spring / Summer /
<code>credits_range</code>	CharField(50)	<i>nullable</i>	Credit range (e.g. 12-18 credits)
<code>cost_estimate</code>	DecimalField(10,2)	<i>nullable</i>	Estimated total cost in USD
<code>application_deadline</code>	DateField()	<i>nullable</i>	Application deadline
<code>max_participants</code>	IntegerField()	<i>nullable</i>	Max students accepted
<code>is_active</code>	BooleanField()	<i>default=True</i>	Program currently offered
<code>created_at</code>	DateTimeField()	<i>auto_now_add=True</i>	Record creation timestamp

11. ResearchOpportunity Model

Undergraduate and graduate research positions led by faculty. UW invests \$1B+ annually in research. Students can browse and apply here.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>opportunity_id</code>	AutoField (PK)	<i>PRIMARY KEY</i>	Unique research opportunity ID
<code>title</code>	CharField(300)	<i>NOT NULL</i>	Research opportunity title
<code>faculty</code>	ForeignKey(Faculty)	<i>ON DELETE SET NULL</i>	Lead faculty researcher FK
<code>department</code>	ForeignKey(Department)	<i>ON DELETE SET NULL</i>	Hosting department FK
<code>description</code>	TextField()	<i>nullable</i>	Full description of the research
<code>level</code>	CharField(30)	<i>choices</i>	Undergraduate / Graduate / Both
<code>is_paid</code>	BooleanField()	<i>default=False</i>	Paid position flag
<code>stipend_amount</code>	DecimalField(8,2)	<i>nullable</i>	Monthly/semester stipend

<code>hours_per_week</code>	<code>IntegerField()</code>	<i>nullable</i>	Expected weekly hours commitment
<code>start_date</code>	<code>DateField()</code>	<i>nullable</i>	Position start date
<code>end_date</code>	<code>DateField()</code>	<i>nullable</i>	Position end date
<code>application_deadline</code>	<code>DateField()</code>	<i>nullable</i>	Deadline to apply
<code>skills_required</code>	<code>TextField()</code>	<i>nullable</i>	Required skills/background
<code>is_active</code>	<code>BooleanField()</code>	<i>default=True</i>	Opportunity currently open
<code>created_at</code>	<code>DateTimeField()</code>	<i>auto_now_add=True</i>	Record creation timestamp

12. Event Model

Campus events including academic, athletic, cultural, and career events. Supports both in-person and virtual formats with registration.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>event_id</code>	<code>AutoField (PK)</code>	<i>PRIMARY KEY</i>	Unique event identifier
<code>title</code>	<code>CharField(300)</code>	<i>NOT NULL</i>	Event title
<code>description</code>	<code>TextField()</code>	<i>nullable</i>	Full event description
<code>event_type</code>	<code>CharField(50)</code>	<i>choices</i>	Academic / Sports / Cultural / Career / Research / Social
<code>organizer</code>	<code>ForeignKey(User)</code>	<i>nullable FK</i>	Organizing user/admin FK
<code>school</code>	<code>ForeignKey(School)</code>	<i>nullable FK</i>	Associated school (if any)
<code>start_datetime</code>	<code>DateTimeField()</code>	<i>NOT NULL</i>	Event start date and time
<code>end_datetime</code>	<code>DateTimeField()</code>	<i>NOT NULL</i>	Event end date and time

<code>location_name</code>	CharField(200)	<i>nullable</i>	Venue name
<code>location_address</code>	CharField(300)	<i>nullable</i>	Full venue address
<code>is_virtual</code>	BooleanField()	<i>default=False</i>	Online/virtual event flag
<code>virtual_link</code>	URLField(500)	<i>nullable</i>	Zoom/Teams/webcast URL
<code>max_attendees</code>	IntegerField()	<i>nullable</i>	Maximum registrants allowed
<code>registration_required</code>	BooleanField()	<i>default=False</i>	
<code>registration_deadline</code>	DateTimeField()	<i>nullable</i>	
<code>banner_image_url</code>	URLField(500)	<i>nullable</i>	Event banner/poster image URL
<code>is_public</code>	BooleanField()	<i>default=True</i>	Public vs restricted event
<code>created_at</code>	DateTimeField()	<i>auto_now_add=True</i>	Record creation timestamp

13. Library Model

Campus library locations (e.g. Memorial Library) with hours, contact info, and available services. Accessible from Student Dashboard.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>library_id</code>	AutoField (PK)	<i>PRIMARY KEY</i>	Unique library identifier
<code>name</code>	CharField(200)	<i>NOT NULL</i>	Library name (e.g. Memorial Library)
<code>location_address</code>	CharField(300)	<i>nullable</i>	Full address/building
<code>contact_phone</code>	CharField(20)	<i>nullable</i>	Library phone number
<code>contact_email</code>	EmailField(254)	<i>nullable</i>	Library email
<code>website_url</code>	URLField(500)	<i>nullable</i>	Library website URL
<code>hours_weekday</code>	CharField(100)	<i>nullable</i>	Weekday operating hours
<code>hours_weekend</code>	CharField(100)	<i>nullable</i>	Weekend operating hours

<code>total_volumes</code>	<code>IntegerField()</code>	<i>nullable</i>	Total physical volumes held
<code>has_study_rooms</code>	<code>BooleanField()</code>	<i>default=False</i>	Study rooms available flag
<code>has_printing</code>	<code>BooleanField()</code>	<i>default=True</i>	Printing services available
<code>is_active</code>	<code>BooleanField()</code>	<i>default=True</i>	Library operational status
<code>created_at</code>	<code>DateTimeField()</code>	<i>auto_now_add=True</i>	Record creation timestamp

14. SportsTeam Model

NCAA Division I Wisconsin Badgers teams and club sports. Tracks season records, coaches, and home venues.

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>team_id</code>	<code>AutoField (PK)</code>	<i>PRIMARY KEY</i>	Unique sports team identifier
<code>name</code>	<code>CharField(200)</code>	<i>NOT NULL</i>	Team name (e.g. Wisconsin Badgers Football)
<code>sport</code>	<code>CharField(100)</code>	<i>NOT NULL</i>	Sport type (e.g. Football, Basketball, Volleyball)
<code>division</code>	<code>CharField(50)</code>	<i>choices</i>	
<code>conference</code>	<code>CharField(100)</code>	<i>nullable</i>	Conference
<code>head_coach</code>	<code>CharField(150)</code>	<i>nullable</i>	Head coach full name

<code>assistant_coaches</code>	<code>JSONField()</code>	<i>default=list</i>	List of assistant coach names
<code>home_venue</code>	<code>CharField(200)</code>	<i>nullable</i>	Home venue
<code>season_start</code>	<code>DateField()</code>	<i>nullable</i>	Current season start date
<code>season_end</code>	<code>DateField()</code>	<i>nullable</i>	Current season end date
<code>wins</code>	<code>IntegerField()</code>	<i>default=0</i>	Current season wins
<code>losses</code>	<code>IntegerField()</code>	<i>default=0</i>	Current season losses
<code>website_url</code>	<code>URLField(500)</code>	<i>nullable</i>	Team official website URL
<code>is_active</code>	<code>BooleanField()</code>	<i>default=True</i>	Team currently active
<code>created_at</code>	<code>DateTimeField()</code>	<i>auto_now_add=True</i>	Record creation timestamp

15. Notification Model

User notifications and alerts for enrollment changes, grade postings, event reminders, and system announcements across all dashboards

Field Name	Django Field Type	DB Column / Notes	Description / Example
<code>notification_id</code>	<code>AutoField (PK)</code>	<i>PRIMARY KEY</i>	Unique notification ID
<code>recipient</code>	<code>ForeignKey(User)</code>	<i>ON DELETE CASCADE</i>	Target user FK
<code>title</code>	<code>CharField(200)</code>	<i>NOT NULL</i>	Notification headline
<code>message</code>	<code>TextField()</code>	<i>NOT NULL</i>	Full notification message
<code>notification_type</code>	<code>CharField(50)</code>	<i>choices</i>	Enrollment / Grade / Event / Alert / Announcement

<code>is_read</code>	<code>BooleanField()</code>	<code>default=False</code>	Read status flag
<code>created_at</code>	<code>DateTimeField()</code>	<code>auto_now_add=True</code>	Timestamp when created
<code>read_at</code>	<code>DateTimeField()</code>	<code>nullable</code>	Timestamp when marked read
<code>action_url</code>	<code>URLField(500)</code>	<code>nullable</code>	link URL for the notification

16 .Guardian

Stores guardian/parent information for a student. Students can have up to two guardians via guardian1 and guardian2 ForeignKey fields on the Student model. This design supports all family structures including mother+mother, father+father, mother+father, single parent, legal guardian, etc.

Field Name	Field Type	Null/Blank	Default	Notes / Choices
first_name	CharField(100)	No/No	—	
last_name	CharField(100)	No/No	—	
gender	CharField(20)	Yes/Yes	—	Male, Female, Trans, LGBTQ+, Other
relationship	CharField(50)	No/No	—	Mother, Father, Grandparent, Legal Guardian, Other
email	EmailField	Yes/Yes	—	
phone_primary	CharField(20)	Yes/Yes	—	Primary contact number
phone_secondary	CharField(20)	Yes/Yes	—	
address_line1	CharField(255)	Yes/Yes	—	
city	CharField(100)	Yes/Yes	—	
state	CharField(50)	Yes/Yes	—	

zip_code	CharField(10)	Yes/Yes	—	
is_emergency_contact	BooleanField	No/No	True	
ssn	OneToOneField → SSNRecord	Yes/Yes	—	Optional SSN linkage

17 . SSNRecord — Social Security Number

Stores encrypted Social Security Number data for Students, Staff, and Guardians. Uses Django's GenericForeignKey for a unified SSN table.

Field Name	Field Type	Null/Blank	Default	Notes / Choices
ssn_encrypted	CharField(256)	No/No	—	Encrypted SSN value (never store plain text)
ssn_last4	CharField(4)	No/No	—	Last 4 digits for display/lookup
ssn_hash	CharField(256)	No/No	—	hash for deduplication
content_type	ForeignKey → ContentType	Yes/Yes	—	Generic FK: Student, Staff, Guardian
object_id	PositiveIntegerField	Yes/Yes	—	Generic FK object reference
verification_status	CharField(20)	No/No	Pending	Pending, Verified, Failed, Exempt
verified_date	DateTimeField	Yes/Yes	—	When SSN was verified
created_at	DateTimeField	No/No	auto_now_add	
updated_at	DateTimeField	No/No	auto_now	
created_by	ForeignKey → Staff	Yes/Yes	—	Who entered the SSN

18 . Firearm Record — Firearm / Gun Registration

Stores firearm/gun registration details for Students and Staff who possess a licensed firearm. Uses a Generic ForeignKey so both Students and Staff can be owners.

Field Name	Field Type	Null/Blank	Default	Notes / Choices
owner_content_type	ForeignKey → ContentType	No/No	—	Generic FK: Student or Staff
owner_object_id	PositiveIntegerField	No/No	—	Owner's PK
gun_name	CharField(150)	No/No	—	Common name / nickname of the firearm
manufacturer	CharField(100)	Yes/Yes	—	
model_name	CharField(100)	No/No	—	e.g. Ak-47
serial_number	CharField(100)	No/No	—	Unique per firearm; encrypted at rest
caliber_type	CharField(50)	Yes/Yes	—	e.g. 9mm, .45 ACP, .223 Rem
firearm_type	CharField(50)	No/No	—	Handgun, Rifle, Shotgun, Other
license_number	CharField(100)	No/No	—	Official carry/ownership license number
license_state	CharField(50)	No/No	—	Issuing state of license
license_expiry	DateField	Yes/Yes	—	Expiration date of license

background_check_date	DateField	Yes/Yes	—	Date of background check clearance
storage_location	CharField(200)	Yes/Yes	—	On-campus storage room/locker if applicable
is_registered_on_campus	BooleanField	No/No	False	Campus registration status
registration_date	DateField	Yes/Yes	—	Date registered with campus security
notes	TextField	Yes/Yes	—	Additional remarks
created_at	DateTimeField	No/No	auto_now_add	
updated_at	DateTimeField	No/No	auto_now	

19. StudentProfile Model

Extended academic profile for students, including GPA, standing, major, and advisor. Linked one-to-one with the User model to store student-specific data.

Field Name	Django Field Type	DB Column / Notes	Description / Example
student_id	AutoField (PK)	PRIMARY KEY	Unique student record identifier
user	OneToOneField(User)	ON DELETE CASCADE	Linked User account FK
student_number	CharField(20)	UNIQUE, NOT NULL	Official student ID number (e.g. 9075432)
enrollment_status	CharField(30)	choices, NOT NULL	Full-time / Part-time / Withdrawn / Graduated

academic_standing	CharField(30)	choices	Good Standing / Probation / Suspension / Honors
class_level	CharField(20)	choices	Freshman / Junior / Senior / Graduate
gpa_cumulative	DecimalField(4,3)	nullable	Cumulative GPA (0.000 – 4.000)
gpa_semester	DecimalField(4,3)	nullable	Current semester GPA
major	ForeignKey(Program)	nullable FK	Primary declared major FK
minor	ForeignKey(Program)	nullable FK	Declared minor FK
advisor	ForeignKey(Faculty)	nullable FK	Assigned academic advisor FK
expected_graduation	DateField()	nullable	Expected graduation date
total_credits_earned	IntegerField()	default=0	Total credit hours earned
transfer_credits	IntegerField()	default=0	Transfer credits accepted
is_international	BooleanField()	default=False	International student flag
visa_type	CharField(30)	nullable	e.g. F-1, J-1, H-4
scholarship_info	JSONField()	default=dict	Scholarships held (name, amount, year)
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

20. StaffProfile Model

Non-faculty university staff profiles (administrative, IT, facilities, advising, etc.) linked to the User model. Separate from Faculty to allow different permission sets.

Field Name	Django Field Type	DB Column / Notes	Description / Example
staff_id	AutoField (PK)	PRIMARY KEY	Unique staff record identifier
user	OneToOneField(User)	ON DELETE CASCADE	Linked User account FK
employee_id	CharField(50)	UNIQUE, NOT NULL	University HR employee ID
department	ForeignKey(Department)	nullable FK	Assigned department FK

job_title	CharField(150)	NOT NULL	Official job title (e.g. Academic Advisor)
employment_type	CharField(30)	choices	Full-time / Part-time / Contract / Temporary
office_location	CharField(200)	nullable	Building and room number
office_phone	CharField(20)	nullable	Direct office phone
supervisor	ForeignKey(StaffProfile)	nullable, self-ref FK	Reporting manager FK
hire_date	DateField()	nullable	Date of hire
contract_end_date	DateField()	nullable	Contract end date (if applicable)
is_active	BooleanField()	default=True	Employment active status
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

21. Building Model

Campus buildings and facilities used for classrooms, offices, labs, and services. Referenced by CourseSection, Faculty, and Department models.

Field Name	Django Field Type	DB Column / Notes	Description / Example
building_id	AutoField (PK)	PRIMARY KEY	Unique building identifier
name	CharField(200)	UNIQUE, NOT NULL	Building name (e.g. Paul Hall)
abbreviation	CharField(20)	nullable	Short code used in scheduling (e.g. VV)
building_type	CharField(50)	choices	Academic / Residential / Administrative / Lab / Athletic

address	CharField(300)	nullable	Street address of the building
latitude	DecimalField(9,6)	nullable	GPS latitude for campus map
longitude	DecimalField(9,6)	nullable	GPS longitude for campus map
floors	IntegerField()	nullable	Number of floors
year_built	IntegerField()	nullable	Year constructed
accessibility_features	JSONField()	default=list	e.g. [elevator, ramp, hearing loop]
has_parking	BooleanField()	default=False	Adjacent parking available
image_url	URLField(500)	nullable	Exterior photo URL
map_url	URLField(500)	nullable	Link to campus map pin
is_active	BooleanField()	default=True	Building in use
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

22. HousingUnit Model

On-campus housing units including residence halls, apartments, and family housing. Used by the Housing Portal for room assignment and availability management.

Field Name	Django Field Type	DB Column / Notes	Description / Example
unit_id	AutoField (PK)	PRIMARY KEY	Unique housing unit identifier
building	ForeignKey(Building)	ON DELETE CASCADE	Parent building FK
room_number	CharField(20)	NOT NULL	Room/unit number (e.g. 312A)
unit_type	CharField(50)	choices	Single / Double / Triple / Suite / Apartment / Family
floor	IntegerField()	nullable	Floor number

capacity	IntegerField()	NOT NULL	Max number of occupants
current_occupancy	IntegerField()	default=0	Current number of residents
monthly_rate	DecimalField(8,2)	nullable	Monthly rent in USD
semester_rate	DecimalField(8,2)	nullable	Per-semester rate in USD
has_private_bathroom	BooleanField()	default=False	Private bathroom flag
is_accessible	BooleanField()	default=False	ADA accessible unit
is_gender_inclusive	BooleanField()	default=False	Gender-inclusive housing
status	CharField(30)	choices	Available / Occupied / Maintenance / Reserved
academic_year	CharField(10)	NOT NULL	e.g. 2025-2026
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

23. HousingApplication Model

Student housing applications for on-campus accommodation. Manages preferences, roommate requests, assignment status, and check-in/check-out records.

Field Name	Django Field Type	DB Column / Notes	Description / Example
application_id	AutoField (PK)	PRIMARY KEY	Unique housing application ID
student	ForeignKey (StudentProfile)	ON DELETE CASCADE	Applicant student FK
academic_year	CharField(10)	NOT NULL	e.g. 2025-2026
semester	CharField(20)	choices	Fall / Spring / Summer / Full Year
preferred_unit_type	CharField(50)	nullable, choices	Preferred room type

roommate_request	ForeignKey(StudentProfile)	nullable FK	Requested roommate FK
housing_unit	ForeignKey(HousingUnit)	nullable FK	Assigned unit FK (null until assigned)
status	CharField(30)	choices	Submitted / Waitlisted / Assigned / Cancelled / Checked-in
special_accommodations	TextField()	nullable	Accessibility or medical housing needs
submitted_at	DateTimeField()	auto_now_add=True	Application submission timestamp
assigned_at	DateTimeField()	nullable	Assignment confirmation timestamp
check_in_date	DateField()	nullable	Scheduled or actual check-in date
check_out_date	DateField()	nullable	Scheduled or actual check-out date
notes	TextField()	nullable	Admin/staff notes

24. Transcript Model

Official academic transcript records per student per semester. Captures final GPA, credit summary, and academic standing at the close of each term.

Field Name	Django Field Type	DB Column / Notes	Description / Example
transcript_id	AutoField (PK)	PRIMARY KEY	Unique transcript record ID
student	ForeignKey(StudentProfile)	ON DELETE CASCADE	Student FK
semester	CharField(20)	choices	Fall / Spring / Summer / Winter

year	IntegerField()	NOT NULL	Academic year (e.g. 2025)
credits_attempted	DecimalField(5,2)	NOT NULL	Credits attempted this semester
credits_earned	DecimalField(5,2)	NOT NULL	Credits earned this semester
semester_gpa	DecimalField(4,3)	NOT NULL	GPA for this semester
cumulative_gpa	DecimalField(4,3)	NOT NULL	Cumulative GPA at end of semester
total_credits_earned	DecimalField(6,2)	NOT NULL	Total credits earned through this term
academic_standing	CharField(30)	choices	Good Standing / Probation / Dean's List / Honors
is_official	BooleanField()	default=False	Official sealed transcript flag
generated_at	DateTimeField()	nullable	Date transcript was generated/sealed
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

25. Grade Model

Individual grade records for each student enrollment. Links to Enrollment and stores letter grade, numeric score, and grading notes per assessment or final.

Field Name	Django Field Type	DB Column / Notes	Description / Example
grade_id	AutoField (PK)	PRIMARY KEY	Unique grade record ID

enrollment	ForeignKey(Enrollment)	ON DELETE CASCADE	Linked enrollment FK
grade_type	CharField(50)	choices	Midterm / Final / Assignment / Quiz / Lab / Project
letter_grade	CharField(5)	nullable	A, A-, B+, B, ... F, I, W, P, NP
numeric_score	DecimalField(5,2)	nullable	Raw score (e.g. 87.50)
max_score	DecimalField(5,2)	nullable	Maximum possible score
weight_percent	DecimalField(5,2)	nullable	Weight of this grade in final (%)
graded_by	ForeignKey(Faculty)	nullable FK	Grading instructor FK
graded_at	DateTimeField()	nullable	Date/time grade was entered
notes	TextField()	nullable	Grader comments or feedback
is_final	BooleanField()	default=False	Final grade flag
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

26. Olympic Model

Tracks student-athletes who are current or former Olympic competitors or national team members. shows sport discipline, Olympic games participation, medals, academic accommodations for training schedules, and sponsorship details

Field Name	Django Field Type	DB Column / Notes	Description / Example
olympic_id	AutoField (PK)	PRIMARY KEY	Unique Olympic record identifier
student	OneToOneField(StudentProfile)	ON DELETE CASCADE	Linked student profile FK
sport	CharField(100)	NOT NULL	Olympic sport (e.g. Swimming, Track)
discipline	CharField(100)	nullable	Specific event/discipline (e.g. 100m Butterfly)
country_represented	CharField(100)	NOT NULL	Country represented in Olympic competition
national_team_member	BooleanField()	default=False	Current national team member flag
olympic_years	JSONField()	default=list	List of Olympic Games years participated
medals_gold	IntegerField()	default=0	Total gold medals won
medals_silver	IntegerField()	default=0	Total silver medals won
medals_bronze	IntegerField()	default=0	Total bronze medals won
world_ranking	IntegerField()	nullable	Current world ranking in discipline
coach_name	CharField(150)	nullable	Primary coach/trainer name
training_hours_week	IntegerField()	nullable	Average weekly training hours
training_location	CharField(200)	nullable	Primary training facility/location
academic_accommodation	TextField()	nullable	Approved academic schedule accommodations

sponsorship_details	JSONField()	default=list	List of sponsors (name, amount, type)
sports_team	ForeignKey(SportsTeam)	nullable FK	Associated campus sports team FK
next_competition_date	DateField()	nullable	Upcoming major competition date
is_active_competitor	BooleanField()	default=True	Currently competing at elite level
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

27. P.hd Model

Field Name	Django Field Type	DB Column / Notes	Description / Example
phd_id	AutoField (PK)	PRIMARY KEY	Unique Ph.D. record identifier
student	OneToOneField(StudentProfile)	ON DELETE CASCADE	Linked student profile FK
program	ForeignKey(Program)	ON DELETE CASCADE	Enrolled Ph.D. program FK
advisor	ForeignKey(Faculty)	nullable FK	Primary dissertation advisor FK
co_advisor	ForeignKey(Faculty)	nullable FK	Co-advisor / second reader FK
research_area	CharField(300)	NOT NULL	Primary research area/topic
dissertation_title	CharField(500)	nullable	Working or final dissertation title
qualifying_exam_status	CharField(30)	choices	Not Taken / Passed / Failed /

qualifying_exam_date	DateField()	nullable	Date qualifying exam was taken
application_status	CharField(30)	choices	Not Submitted / Submitted / Approved
application_date	DateField()	nullable	Prospectus approval date
committee_members	ManyToManyField(Faculty)	blank=True	Dissertation committee members
funding_type	CharField(50)	choices	Fellowship / Self / Other
funding_amount	DecimalField(10,2)	nullable	Annual funding amount in USD
funding_source	CharField(200)	nullable	Funding source name or grant ID
years_in_program	IntegerField()	default=0	Number of years in Ph.D. program
expected_defense_date	DateField()	nullable	Planned dissertation defense date
actual_defense_date	DateField()	nullable	Actual defense date (if completed)
defense_outcome	CharField(30)	nullable, choices	Pass / Pass with Revisions / Fail
graduation_date	DateField()	nullable	Official graduation/conferral date
publications	JSONField()	default=list	List of published papers (title, journal, year)
is_active	BooleanField()	default=True	Currently active candidate
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

28. FreeLance Model

Field Name	Django Field Type	DB Column / Notes	Description / Example
freelance_id	AutoField (PK)	PRIMARY KEY	Unique freelance record identifier
user	ForeignKey(User)	nullable FK	Linked User account (if registered)
full_name	CharField(200)	NOT NULL	Freelancer full name
email	EmailField(254)	UNIQUE, NOT NULL	Primary contact email
phone_number	CharField(20)	nullable	Contact phone number
expertise_area	CharField(300)	NOT NULL	e.g. Web Development, Graphic Design
skills	JSONField()	default=list	List of skills (e.g. [Python, React])
portfolio_url	URLField(500)	nullable	Link to portfolio or LinkedIn
contract_type	CharField(30)	choices	Hourly / Fixed / Project-based / Retainer
hourly_rate	DecimalField(8,2)	nullable	Hourly billing rate in USD
project_rate	DecimalField(10,2)	nullable	Fixed project rate in USD
contract_start_date	DateField()	nullable	Contract start date
contract_end_date	DateField()	nullable	Contract end/expiry date
department	ForeignKey(Department)	nullable FK	Assigned department FK
project_title	CharField(300)	nullable	Title of the assigned project
deliverables	JSONField()	default=list	List of expected deliverables with deadlines
payment_method	CharField(50)	choices	Bank Transfer / Check / Other
tax_id	CharField(50)	nullable	Tax ID / SSN for IRS Form 1099

is_active	BooleanField()	default=True	Currently active contract status
rating	DecimalField(3,2)	nullable	Internal rating score (0.00 – 5.00)
notes	TextField()	nullable	Admin/department internal notes
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

29. Quota Model

Field Name	Django Field Type	DB Column / Notes	Description / Example
quota_id	AutoField (PK)	PRIMARY KEY	Unique quota record identifier
quota_name	CharField(200)	NOT NULL	Descriptive quota name (e.g. Int'l Seats – CS)
quota_type	CharField(50)	choices, NOT NULL	Course / Program / Housing / Department
target_model	CharField(50)	choices, NOT NULL	Course / Program / HousingUnit / Department
target_id	IntegerField()	NOT NULL	Primary key of the target object
category	CharField(50)	choices	General / International / Transfer / Graduate / Scholarship
total_seats	IntegerField()	NOT NULL	Total seats allocated under this quota
reserved_seats	IntegerField()	default=0	Seats reserved for specific category
occupied_seats	IntegerField()	default=0	Currently filled/used seats
waitlisted_count	IntegerField()	default=0	Number of users on waitlist
min_gpa_required	DecimalField(4,3)	nullable	Minimum GPA to qualify for reserved seats
eligibility_criteria	JSONField()	default=dict	Key-value eligibility rules (role, status, etc.)
semester	CharField(20)	choices	Fall / Spring / Summer / Full Year

academic_year	CharField(10)	NOT NULL	e.g. 2025-2026
open_date	DateTimeField()	nullable	Date/time quota opens for enrollment
close_date	DateTimeField()	nullable	Date/time quota closes
auto_waitlist	BooleanField()	default=True	Auto-enroll overflow into waitlist
overflow_limit	IntegerField()	nullable	Max waitlist size before hard close
department	ForeignKey(Department)	nullable FK	Owning department FK (if applicable)
notes	TextField()	nullable	Admin notes or quota policy description
is_active	BooleanField()	default=True	Quota currently in effect
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

30. VIP Model

Field Name	Django Field Type	DB Column / Notes	Description / Example
vip_id	AutoField (PK)	PRIMARY KEY	Unique VIP record identifier
user	OneToOneField(User)	ON DELETE CASCADE	Linked User account FK
vip_code	CharField(50)	UNIQUE, NOT NULL	Unique VIP identifier code
vip_category	CharField(50)	choices, NOT NULL	Donor / Alumni / Board Member / Government / Partner
tier	CharField(20)	choices, NOT NULL	Gold / Platinum / Diamond
designation_reason	TextField()	nullable	Reason for VIP designation

relationship_manager	ForeignKey(StaffProfile)	nullable FK	Assigned relationship manager FK
organization	CharField(200)	nullable	Affiliated organization or company
title	CharField(100)	nullable	Official title (e.g. Senator, CEO, Dean Emeritus)
donation_total	DecimalField(12,2)	default=0.00	Cumulative donation amount in USD
last_donation_date	DateField()	nullable	Date of most recent donation
event_invitations	JSONField()	default=list	List of events with exclusive invitations
access_privileges	JSONField()	default=list	Granted platform access privileges
preferred_contact	CharField(30)	choices	Email / Phone / In-Person / None
special_requests	TextField()	nullable	Notes on special service requirements
badge_url	URLField(500)	nullable	URL to VIP badge/icon image
is_active	BooleanField()	default=True	VIP status active
vip_since	DateField()	nullable	Date VIP status was first granted
expiry_date	DateField()	nullable	VIP status expiry date (null = indefinite)
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp

31. VVIP Model

Field Name	Django Field Type	DB Column / Notes	Description / Example
vvip_id	AutoField (PK)	PRIMARY KEY	Unique VVIP record identifier

vip_profile	OneToOneField(VIP)	ON DELETE CASCADE	Linked VIP profile FK
vvip_code	CharField(50)	UNIQUE, NOT NULL	Unique VVIP identifier code
vvip_category	CharField(50)	choices, NOT NULL	Chancellor(head of UW) / Philanthropist(donates money, resources,) / Head of State / Strategic Partner(Business collaboration)
protocol_officer	ForeignKey(StaffProfile)	nullable FK	Assigned protocol/ Guest Relations Officer FK
security_clearance	BooleanField()	default=False	Security clearance required flag
security_notes	TextField()	nullable	Confidential security & protocol notes
private_event_access	JSONField()	default=list	List of exclusive private event IDs
dedicated_relaxing_area	BooleanField()	default=False	Access to dedicated VIP relaxing area e/suite
Guest Relations_service	BooleanField()	default=True	24/7 concierge service enabled
personal_aide	CharField(200)	nullable	Personal aide or chief of staff name
lifetime_giving	DecimalField(14,2)	default=0.00	Total lifetime giving to the university (USD)
donation_name	CharField(300)	nullable	Named donation_name or building (if applicable)
special_degree	BooleanField()	default=False	special_degree conferred flag
special_degree_year	IntegerField()	nullable	Year special_degree was conferred
board_seat	BooleanField()	default=False	Holds a seat on Board of Regents
board_seat_since	DateField()	nullable	Date board seat was assumed
travel_preferences	JSONField()	default=dict	Arrival/departure logistics preferences
dietary_requirements	JSONField()	default=list	Special dietary needs for events

media_restrictions	TextField()	nullable	Photography/media blackout instructions
nda_on_file	BooleanField()	default=False	Non-disclosure agreement on file
is_active	BooleanField()	default=True	VVIP status currently active
vvip_since	DateField()	nullable	Date VVIP status was first granted
created_at	DateTimeField()	auto_now_add=True	Record creation timestamp